

CO3-ASSESSMENT TOOL 2 – CASE STUDY ANSWERS

Course Code: ITA0502 Course Name: Computer Vision
Name: Konjeti Pavan Deekshith
Reg No: 192412172

Question 1

A remote sensing application is used to align satellite images of the same geographical region captured at different times for monitoring environmental changes. The system initially relies on edge detection to identify matching regions between the images; however, due to differences in illumination, image orientation, and background details, the matching process becomes unreliable and produces inaccurate alignment results. The development team decides to replace the edge-based approach with Harris Corner Detection to identify stable and distinctive feature points. Analyze the limitations of edge detection in this application and explain how Harris Corner Detection improves feature localization, matching accuracy, and robustness for image registration under varying imaging conditions.

1. Understanding of the Case

The case describes a multi-temporal satellite image registration problem, where the goal is to align two or more images of the same geographic area captured at different times, so that pixel-level or region-level changes (e.g., deforestation, urban growth, flooding) can be reliably monitored. The original system used edge detection as its feature extraction technique to identify matching regions between images. Edge-based matching failed to produce accurate alignment because satellite images captured at different times differ in illumination (sun angle, season, atmospheric conditions), orientation (satellite viewing angle, image rotation), and background clutter (clouds, vegetation changes, seasonal variation). These factors change the location, strength, and continuity of edges between the two images, even though the underlying geographic features have not moved. This is the central problem the case is testing: recognizing that edges, while useful for outlining object boundaries, are not always distinctive or stable enough to serve as reliable correspondence points.

2. Limitations of Edge Detection in This Application

- **Sensitivity to illumination changes:** Edge strength and even the presence of an edge depend heavily on local contrast. Since satellite images of the same region taken months or years apart have different sun angles and shadow patterns, the same physical boundary (e.g., a field edge or road) may produce a strong edge in one image and a weak or missing edge in the other, causing false or missed correspondence.
- **Lack of distinctiveness:** An edge point is only locally distinctive along one dimension — it looks similar to every other point along the same edge line (this is known as the aperture problem). Two different points on the same straight edge cannot be reliably told apart, so the matching algorithm can associate the wrong points across images, producing misalignment.

- **Sensitivity to orientation and rotation:** Edge maps are directional; if one satellite pass is rotated relative to another, the detected edge orientations and connectivity patterns change substantially, making point-to-point correspondence unreliable without expensive re-alignment.
- **Vulnerability to background clutter and noise:** Clouds, seasonal vegetation, and sensor noise generate spurious or broken edges that do not correspond to real ground features, adding false matches and reducing registration accuracy.
- **No inherent "corneriness" or repeatability guarantee:** Edge detectors (e.g., Sobel, Canny) return continuous, one-dimensional contours rather than well-localized, repeatable interest points, so the same edge pixel is unlikely to be re-detected at the same location in a slightly transformed image.

3. How Harris Corner Detection Improves the System

Harris Corner Detection identifies points where the image intensity changes significantly in two orthogonal directions simultaneously (i.e., true corners, not just edges). It does this by analyzing the local structure tensor (second-moment matrix) of image gradients within a small window and computing a response function R based on the eigenvalues of this matrix. Points where both eigenvalues are large — indicating strong intensity variation in all directions — are flagged as corners.

- **Improved feature localization:** Because a corner is defined by intensity change in two directions, it pinpoints an exact (x, y) location rather than a line, eliminating the aperture problem that affects edge points. This gives sub-pixel-accurate, spatially unique landmarks such as building corners, road intersections, and field boundaries.
- **Higher matching accuracy:** Corner points are locally distinctive, so a small image patch around a Harris corner has a unique intensity pattern that can be reliably matched to the corresponding patch in the second image, drastically reducing false correspondences compared to matching along ambiguous edge contours.
- **Robustness to illumination changes:** The Harris response is derived from gradient direction and relative intensity variation rather than absolute intensity, making it largely invariant to uniform and moderate illumination changes — a key requirement given the seasonal and time-of-day differences in satellite captures.
- **Rotation invariance:** The eigenvalues of the second-moment matrix are invariant under image rotation, so corners detected in a rotated satellite pass still correspond geometrically to the same physical points, enabling reliable matching even when successive images are not perfectly aligned in orientation.
- **Robustness to background clutter:** Since Harris corners require strong, structured gradients in two directions, random noise, cloud texture, or smooth vegetation regions rarely trigger high corner responses, naturally suppressing spurious matches that plagued the edge-based approach.

- **Repeatability across viewpoints:** Harris corners exhibit high repeatability — the same physical corner tends to be re-detected under moderate scale, rotation, and viewpoint changes, which is essential when the same geographic region is imaged from slightly different satellite passes over time.

Figure 1: Edge Detection vs Harris Corner Detection for Feature-Based Image Registration

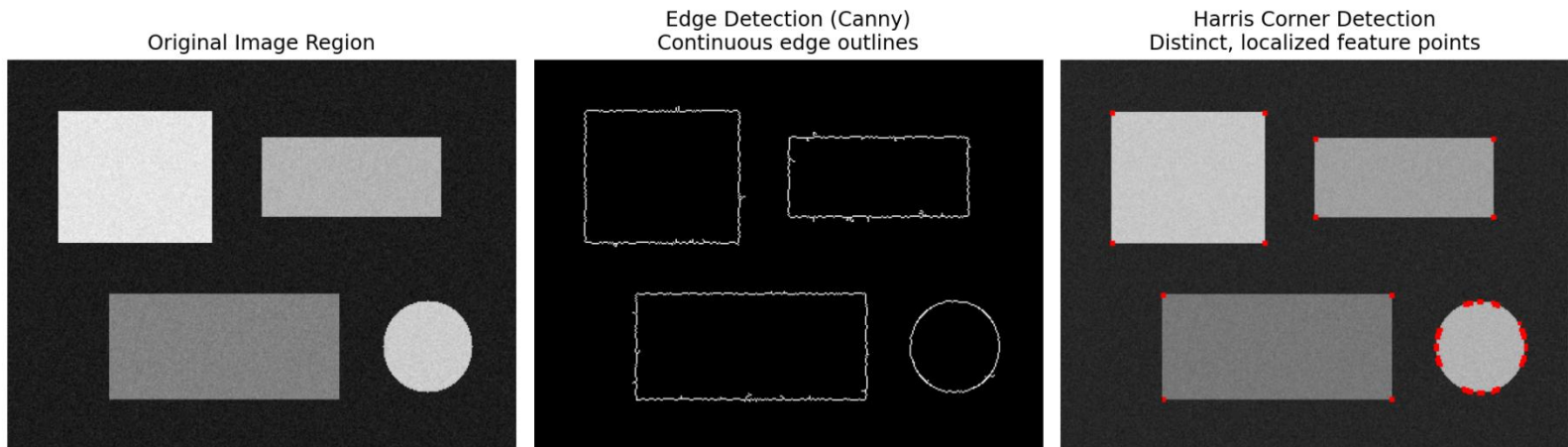


Figure 1: Edge detection produces continuous but ambiguous contours, whereas Harris Corner Detection yields sparse, well-localized, and distinctive feature points suitable for reliable image registration.

4. Conclusion / Solution Justification

Replacing edge detection with Harris Corner Detection directly addresses the root cause of the registration failures — the lack of spatially unique, illumination- and rotation-invariant feature points. By extracting well-localized corner features, the remote sensing system can compute an accurate geometric transformation (e.g., using RANSAC-based homography estimation on matched corner pairs) between images captured at different times, resulting in precise alignment and dependable environmental change detection. For further robustness against scale changes (e.g., different satellite altitudes or zoom levels), the team could additionally consider scale-invariant descriptors such as SIFT, which builds on the same principle of distinctive key point localization.

Question 2

An autonomous surveillance system continuously captures high-resolution color images to detect vehicles and pedestrians in real time. Processing the original color images directly increases computational complexity and slows down feature extraction, making it difficult to achieve the required processing speed. To improve system performance, the engineers consider converting the images into grayscale or binary representations before applying feature detection algorithms. Analyze the impact of different image representations on computational efficiency, edge detection, and feature extraction performance, and justify the most appropriate image representation for achieving accurate and reliable object detection in real-time computer vision applications.

1. Understanding of the Case

The surveillance system needs to detect vehicles and pedestrians in real time from continuously captured high-resolution color videos. The bottleneck identified is computational: processing full-color (RGB) images directly makes feature extraction too slow to meet real-time constraints. The case asks us to evaluate three candidate image representations — color (RGB), grayscale, and binary — in terms of computational efficiency, their effect on edge detection, and their effect on feature extraction/object detection performance and then justify which representation (or combination) is most appropriate.

2. Impact of Image Representation on Computational Efficiency

- **Color (RGB) images:** Store three channels (Red, Green, Blue) per pixel, tripling the raw data volume compared to a single-channel image. Every filtering, convolution, or gradient operation used in feature detection (e.g., Sobel, Harris, HOG) must, in principle, be computed per channel or on a combined representation, increasing memory bandwidth usage and processing time — a critical problem for high-resolution, real-time video streams.
- **Grayscale images:** Reduce the data to a single intensity channel, cutting the data volume roughly to one-third of the RGB version. This significantly speeds up convolution-based operations (edge detection, corner detection, gradient computation) since only one channel needs to be processed, making grayscale a common first step in real-time pipelines.
- **Binary images:** Represent each pixel using a single bit (foreground/background, 0 or 1) after thresholding or background subtraction. This is the most memory- and computation-efficient representation, enabling extremely fast operations such as connected-component analysis, contour extraction, and blob detection, but it is only obtained after discarding most of the original intensity/texture information.

3. Impact on Edge Detection

- **Color images:** Edge detection can technically be run per channel and combined, but this increases computation roughly three-fold and can produce inconsistent edges across channels (an edge visible in the red channel may be weak in blue), complicating the result without a proportional gain in accuracy for most surveillance tasks.
- **Grayscale images:** Provide a single, well-defined intensity gradient field, so standard edge detectors (Sobel, Canny) run efficiently and produce a clean, consistent edge map that captures object boundaries (vehicle outlines, pedestrian silhouettes) needed for shape-based detection.
- **Binary images:** Edge detection is reduced to finding the boundary between foreground and background regions (contour extraction), which is very fast but only useful if the prior thresholding/segmentation step has already reliably separated the object from the background — it discards internal texture edges that may help distinguish, for example, a pedestrian from a similarly shaped pole.

4. Impact on Feature Extraction and Detection Performance

- **Color images:** Retain the most information, including color cues (e.g., distinguishing a red car from the grey road, or skin/clothing tones for pedestrians), which can improve detection accuracy in cluttered scenes, but at a high computational cost that conflicts with the real-time requirement.
- **Grayscale images:** Preserve shape, texture, and intensity-gradient information, which is what most classical feature descriptors (HOG, Harris corners, SIFT-like gradients) and even many deep learning backbones primarily rely on for detecting object shape and structure. This makes grayscale a strong balance between accuracy and speed for vehicle/pedestrian shape detection.
- **Binary images:** Are excellent for fast motion/foreground detection (e.g., background subtraction + thresholding to instantly flag moving blobs), but lose fine texture and intensity gradient detail, making them unreliable on their own for distinguishing between object classes (vehicle vs. pedestrian vs. shadow) or handling partial occlusion and complex backgrounds.

Figure 2: Impact of Image Representation on Data Volume and Feature Extraction

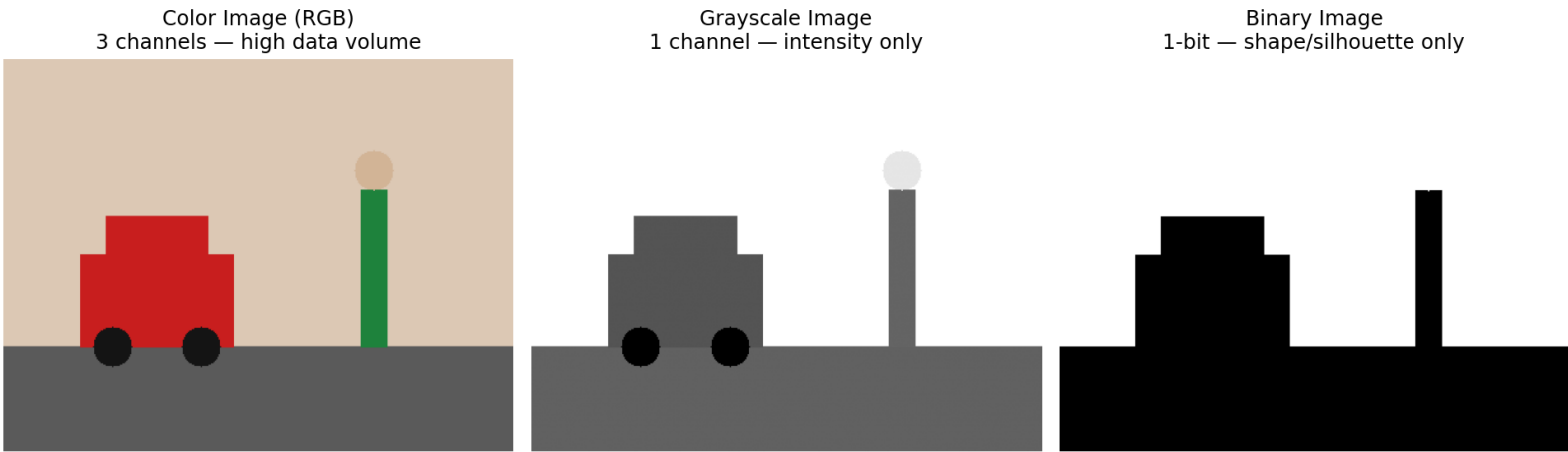


Figure 2: Color images carry the most information but are computationally heaviest; grayscale retains shape and gradient detail at one-third the data volume; binary images are fastest but preserve only silhouette information.

5. Comparative Summary

Aspect	Color (RGB)	Grayscale	Binary
Data volume	High (3 channels)	Medium (1 channel)	Very low (1 bit/pixel)
Processing speed	Slowest	Fast	Fastest
Edge/shape detail retained	Full detail + color cues	Full shape & gradient detail	Silhouette/contour only
Robustness to clutter/occlusion	Good (with high cost)	Good	Weak (loses texture)
Best suited for	Offline / high-accuracy analysis	Real-time shape-based object detection	Fast motion/foreground flagging

6. Justification of the Most Appropriate Representation

For an autonomous surveillance system that must detect vehicles and pedestrians in real time, grayscale representation is the most appropriate primary choice for feature detection. It reduces data volume and computation by roughly two-thirds compared to color, directly addressing the stated performance bottleneck, while still preserving the intensity gradients and shape information that classical and learning-based detectors depend on to recognize object structure (silhouette, edges, corners) — information that a binary image would irreversibly discard.

In practice, the most effective and commonly used real-time pipeline combines representations in stages rather than relying on a single one: (1) a fast binary foreground mask (via background subtraction/thresholding) is used first to quickly localize candidate moving regions and eliminate most of the static background at very low cost; (2) grayscale conversion and feature extraction (e.g., HOG, Harris/edge-based descriptors, or a lightweight CNN) is then applied only within those candidate regions

to classify and confirm whether the object is a vehicle or pedestrian; and (3) color information is retained only as an optional secondary cue (e.g., for logging, re-identification, or disambiguating objects in low-contrast scenes) rather than as the primary channel for detection.

This staged approach achieves the required real-time processing speed by minimizing the area and channel depth over which expensive feature extraction is performed, while still retaining enough intensity and shape information from the grayscale stage to achieve accurate and reliable vehicle/pedestrian detection — directly satisfying the case's dual requirement of computational efficiency and detection accuracy.